Loan Performance & DEFAULT ANALYTICS



Project objective

- To analyze historical loan portfolio data (2013–2018) in order to identify patterns and key drivers of loan default, and to provide stakeholders with an interactive Power BI dashboard for monitoring risk across borrower demographics, credit profiles, and financial segments.



Data Source - DataFlow

Downloading, Installing & Configuring Standard Mode Gateway

Downloading & Installing Microsoft SQL Server

Importing Data to SQL Server

Creating the Dataflow using Power BI Service

Importing Data into Power BI Desktop from dataflow

DAX Queries

1. **Year** = YEAR('Loan_default'[Loan_Date_DD_MM_YYYY].[Date])

2. **Income Bracket** = SWITCH(
TRUE(),
'Loan_default'[Income]<30000,"Low Income"
'Loan_default'[Income]>=30000 && 'Loan_default'[Income]<60000,"Medium Income",
'Loan_default'[Income]>=60000, "High Income")

3. **Credit Score Bins** =
IF(Loan_default[CreditScore]<=400,"Very Low",
IF('Loan_default'[CreditScore]<=450,"Low",
IF('Loan_default'[CreditScore]<=650,"Medium","High")))

4. **Age Groups** = IF('Loan_default'[Age]<=19,"Teen",
IF('Loan_default'[Age]<=39,"Adults",
IF('Loan_default'[Age]<=59,"Middle Age Adults","Senior Citizens")))

5. **Loans by Education type** = COUNTROWS(FILTER('Loan_default',NOT(ISBLANK('Loan_default'[LoanID]))))

6. **Average Loan Amt (High Credit)** =
AVERAGEX(FILTER('Loan_default','Loan_default'[Credit Score. Bins]="High"),'Loan_default'[LoanAmount])

7. **YTD Loan Amount** =
CALCULATE(SUM('Loan_default'[LoanAmount]),DATESYTD('Loan_default'[Loan_Date_DD_MM_YYYY]
[Date]),ALLEXCEPT('Loan_default',Loan_default[Credit Score Bins],Loan_default[MaritalStatus]))

DAX QUERIES

8. YOY Default Loans change =
DIVIDE(CALCULATE(COUNTROWS(FILTER('Loan_default','Loan_default'[Default]=TRUE())),'Loan_default'[Year]=YEAR(MAX('Loan_default'[Loan_Date_DD_MM_YYYY]))) -

CALCULATE(COUNTROWS(FILTER('Loan_default','Loan_default'[Default]=TRUE())),'Loan_default'[Year]=YEAR(MAX('Loan_default'[Loan_Date_DD_MM_YYYY]))-1)

,CALCULATE(COUNTROWS(FILTER('Loan_default','Loan_default'[Default]=TRUE())),'Loan_default'[Year]=YEAR(MAX('Loan_default'[Loan_Date_DD_MM_YYYY]))-1),0) * 100

9. YOY Loan Amount Change =
DIVIDE(
CALCULATE(SUM('Loan_default'[LoanAmount]),'Loan_default'[Year]=YEAR(MAX('Loan_default'[Loan_Date_DD_MM_YYYY]))) -
CALCULATE(SUM('Loan_default'[LoanAmount]),'Loan_default'[Year]=YEAR(MAX('Loan_default'[Loan_Date_DD_MM_YYYY]))-1),
CALCULATE(SUM('Loan_default'[LoanAmount]),'Loan_default'[Year]=YEAR(MAX('Loan_default'[Loan_Date_DD_MM_YYYY]))-1),0)
* 100

10. Median by Credit score bins = MEDIANX('Loan_default','Loan_default'[LoanAmount])

11. Total Loan (Middle Age Adults) =
SUMX(FILTER('Loan_default','Loan_default'[Age Groups]="Middle Age. Adults"),'Loan_default'[LoanAmount])

DAX QUERIES

12. Total Loan (Credit Bins) =

```
CALCULATE(SUM('Loan_default'[LoanAmount]),'Loan_default'[Age Groups]="Adults",ALLEXCEPT('Loan_default','Loan_default'[Age],'Loan_default'[Age Groups],'Loan_default'[CreditScore],'Loan_default'[Credit Score Bins]))
```

13. Average Income by Employment type =

```
CALCULATE(AVERAGE('Loan_default'[Income]),ALLEXCEPT('Loan_default','Loan_default'[EmploymentType]))
```

14. Average Loan by Age Group =

```
AVERAGEX(VALUES('Loan_default'[Age Groups]),AVERAGE('Loan_default'[LoanAmount]))
```

15. Default Rate by Employment type =

```
Var totalrecords = COUNTROWS(ALL('Loan_default'))
Var DefaultCases = COUNTROWS(FILTER('Loan_default','Loan_default'[Default]=TRUE()))
RETURN
CALCULATE(DIVIDE(DefaultCases,totalrecords),ALLEXCEPT('Loan_default','Loan_default'[EmploymentType])) *100
```

16. Default Rate by Year =

```
Var totalloans = CALCULATE(COUNTROWS('Loan_default'),ALLEXCEPT('Loan_default',Loan_default[Year]))
Var default =
CALCULATE(COUNTROWS(FILTER('Loan_default','Loan_default'[Default]=TRUE())),ALLEXCEPT('Loan_default',Loan_default[Year]))
RETURN
DIVIDE(default,totalloans)*100
```

17. Loan Amount by Purpose =

```
SUMX(FILTER('Loan_default',NOT(ISBLANK('Loan_default'[LoanAmount]))),'Loan_default'[LoanAmount])
```

Project Insights

📊 **Loan Volume & Purpose**

- Total loan amount is fairly evenly spread across purposes:. Home 6,545M, Auto 6,522M, Education 6,511M, Other 6,501M, Business 6,498M — a spread of less than 1% between highest and lowest.

👥 **Default Rate by Employment Type**

- Unemployed: 3.39% (highest risk segment)
- Full-time: 3.01%
- Self-employed: 2.86%
- Part-time: 2.36% (lowest)
- Gap between highest and lowest default rate: ~1.03 percentage points

💰 **Average Income by Employment Type**

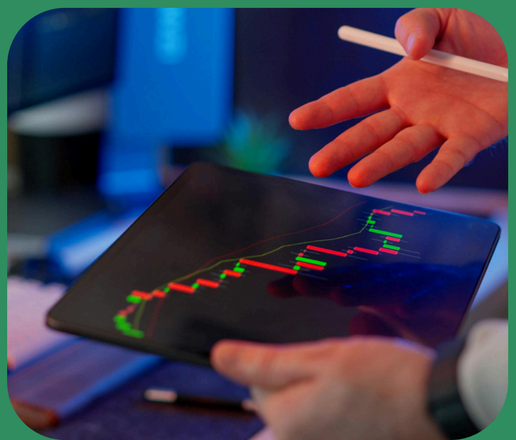
- Full-time: 82,890, Unemployed: 82,447, Part-time: 82,389, Self-employed: 82,272
- Income is nearly flat across groups (<1% variance), suggesting default rate differences aren't purely income-driven — employment stability likely matters more.

📈 **Default Rate Trend (2013–2018)**

- 2013: 11.62% → 2014: 11.50% → 2015: 11.70% (Low) → 2016: 11.75%(peak) → 2017: 11.50% → 2018: 11.60%
- Overall range: 11.50%–11.75%, a tight 0.25 percentage point band — indicating a stable, not worsening, default environment over 6 years.

🔄 **YoY Loan Amount Change**

- 2018: +1.73%, 2014: -1.53%, 2015: +1.30%, 2017: -1.08%, 2016: -0.01%
- Volume growth is essentially flat over the period (oscillating, no sustained upward or downward trend).



Project Insights

🔄 **YoY Default Loans Change**

- 2018: +1.9%, 2015: +2.7%, 2017 -2.8%, 2014: -2.6%, 2016: +0.8%
- Two consecutive years of decline (2014–2017) followed by a rebound in 2018 — worth flagging as a recovery-risk signal for portfolio monitoring.

🏦 **Credit Score Segments (Adults, Total Loan)**

- Medium: 4.6bn (largest),High : 4.5bn, Very Low: 2.3bn, Low: 1.1bn (smallest)

💵 **Loan Amount by Income Bracket**

- High Income: 21.73bn (66.7%)
- Medium Income: 7.21bn (22.1%)
- Low Income: 3.63bn (11.1%)
- Total: 32.58bn

👔 **Loan Amount by Employment Type (absolute)**

- Full-time: 5.44bn, Part-time: 5.44bn, Self-employed: 5.43bn, Unemployed: 5.42bn — near-identical distribution (~25% each), meaning risk exposure isn't concentrated by employment type volume, only by default rate.

🎓 **Loans by Education Type**

- Bachelor's: 64,366, High School : 63,903, Masters : 63,541, PhD: 63,537 — evenly distributed (~25% each), no strong education-based skew in loan count.

